The quasi-Riemann hypothesis

A sketch of the proof

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The Riemann zeta function

Let

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}, \qquad \operatorname{Re} s > 1.$$

Introduced by Euler.

Main insight: by unique factorization

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}$$

with p over primes.

Infinitely many primes

In particular, since

$$\sum_{n \geq 1} \frac{1}{n^s} \to \infty \quad \text{as } s \to 1^+,$$

we have

$$\prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \to \infty \quad \text{as } s \to 1^+,$$

and there are infinitely many primes.

Riemann

Analytically continued by Riemann to all of $\mathbb{C}$ (simple pole at $s = 1$).

Who also showed that if

$$\xi(s) := \pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\zeta(s)$$

then

$$\xi(s) = \xi(1-s).$$

The functional equation

This property, namely $\xi(s) = \xi(1-s)$, is (essentially) equivalent to the Poisson summation formula

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \widehat{f}(n),$$

which captures that the integers form a lattice.

Two main properties

The Euler product

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}$$

and the functional equation

$$\xi(s) = \xi(1-s)$$

are the 2 main properties of $\zeta(s)$.

They reflect respectively unique factorization and that the integers form a lattice.

Primes and zeros

$\zeta(s)$ now has 2 product representations

$$\prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \quad \text{and} \quad \prod_\rho \left(1 - \frac{s}{\rho}\right) \quad \text{over zeros (up to simple factors).}$$

Equating these two and taking logarithmic derivatives we get

$$\sum_p \sum_{k \geq 1} \frac{\log p}{p^{ks}} = \frac{1}{s-1} - \sum_\rho \frac{1}{s-\rho} + \dots$$

Integrating against a suitable kernel we get

$$\sum_{p \leq X} \log p = X - \sum_\rho \frac{X^\rho}{\rho} + O(\sqrt{X}).$$

Zeros control primes

We then see that if $\beta = \sup \operatorname{Re} \rho$ over the zeros ρ of zeta, then

$$\sum_{p \leq X} \log p = X + O(X^{\beta} \log^2 X).$$

The result of OpenAI is that

$$\beta \leq \frac{11}{12}.$$

The Möbius function

Another closely related object is the Möbius function, defined as

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n = p_1 \cdots p_k \text{ (distinct primes)}, \\ 0 & \text{if } p^2 \mid n. \end{cases}$$

This is the coefficient of

$$\frac{1}{\zeta(s)} = \prod_p \left(1 - \frac{1}{p^s}\right) = \sum_{n \geq 1} \frac{\mu(n)}{n^s}.$$

Möbius and zeros

In particular if

$$M(x) = \sum_{n \le x} \mu(n) \ll x^{1-\delta}$$

then since

$$\frac{1}{\zeta(s)} = s \int_1^\infty \frac{M(x)}{x^{s+1}}\, dx$$

we get that $1/\zeta(s)$ is analytic in $\operatorname{Re} s > 1 - \delta$,

and thus $\zeta(s) \neq 0$ in $\operatorname{Re} s > 1 - \delta$.

The Eisenstein integers

Let's now explain the proof. For this we need to take a look at

$$\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}, \qquad \omega = \frac{1 + \sqrt{-3}}{2}.$$

These are known as the Eisenstein integers.

- Addition: they form a lattice
- Units: $\pm 1, \pm\omega, \pm(1 - \omega)$ (the six sixth roots of unity)
- Unique factorization domain

The Dedekind zeta function

Thus $\mathbb{Z}[\omega]$ is analogous to $\mathbb{Z}$, and we should be able to construct an associated ζ function.

Called the Dedekind zeta function of $\mathbb{Z}[\omega]$:

$$\zeta_{\mathbb{Z}[\omega]}(s) = \sum_{\substack{0 \neq n \in \mathbb{Z}[\omega] \\ \text{up to units}}} \frac{1}{|n|^{2s}}.$$

Nothing fundamentally new

$\zeta_{\mathbb{Z}[\omega]}(s)$ is not a fundamentally new object, and factors as

$$\zeta_{\mathbb{Z}[\omega]}(s) = \zeta(s)\, L(s, \chi_{-3}).$$

In particular, if $\zeta_{\mathbb{Z}[\omega]}(s) \neq 0$ then $\zeta(s) \neq 0$.

Everything transfers

Essentially all of the properties transfer:

- Functional equation
- Euler product
- Explicit formula
- Möbius function

Moreover, the number of $n \in \mathbb{Z}[\omega]$ with $|n|^2 \leq X$ is $\approx X$.

Möbius on $\mathbb{Z}[\omega]$

In particular $\mu_{\mathbb{Z}[\omega]} : \mathbb{Z}[\omega] \to \{-1, 0, 1\}$ is defined by

$$\mu_{\mathbb{Z}[\omega]}(n) = \begin{cases} (-1)^k & \text{if } n = \pi_1 \pi_2 \cdots \pi_k \text{ (distinct primes)}, \\ 0 & \text{if } \pi^2 \mid n, \end{cases}$$

and if

$$\sum_{|n|^2 \leq X} \mu_{\mathbb{Z}[\omega]}(n) \ll X^{1-\delta}$$

then $\zeta_{\mathbb{Z}[\omega]}(s) \neq 0$ for $\operatorname{Re} s > 1 - \delta$. Thus also $\zeta(s) \neq 0$ for $\operatorname{Re} s > 1 - \delta$.

From now on everything will be an Eisenstein integer, so we write $\mu(n)$ to mean $\mu_{\mathbb{Z}[\omega]}(n)$.

Amplification

How can one bound

$$(*) \qquad \sum_{|n|^2 \leq X} \mu(n) \ ?$$

A standard technique (usually applied to "smooth" coefficients!) is amplification.

This bounds $(*)$ by an average of "twisted" expressions, so that one can benefit from harmonic analysis.

A first example

Here is a first example:

$$\Big|\sum_{|n|^2\le N}\mu(n)\Big|^2 \ll \log N\int_{\mathbb{R}}\Big|\sum_{|n|^2\le N}\mu(n)|n|^{2it}\Big|^2\Phi\Big(\frac{t}{T}\Big)\,dt.$$

Picking $T = N/H$ and expanding, we get that the RHS is (up to $\log N$)

$$\approx \frac{N^2}{H} + \frac{N}{H}\sum_{\substack{m\neq n\\ ||m|^2-|n|^2|\le H}}\mu(m)\mu(n).$$

Thus we recover Weyl differencing (in van der Corput's form). But these correlations of μ are Chowla-type: harder than what we started with.

Arithmetic twists

But nothing forces us to twist by $|n|^{2it}$, and we can instead twist by different, perhaps more arithmetic, objects.

An appealing twist: cubic, quadratic and sextic characters. These are defined, for π prime ($\pi \nmid 6a$), by the congruence

$$\left(\frac{a}{\pi}\right)_k \equiv a^{\frac{|\pi|^2-1}{k}} \pmod{\pi}.$$

In effect $\left(\frac{a}{\pi}\right)_k$ is

- ± 1 for $k = 2$,
- a cube root of unity for $k = 3$, a sixth root of unity for $k = 6$,
- and does not exist for other values of k.

Residue symbols

- Moreover, for composite b, we define $\left(\frac{a}{b}\right)_6$ by writing $b = \pi_1 \cdots \pi_k$ and setting

$$\left(\frac{a}{b}\right)_6 = \left(\frac{a}{\pi_1}\right)_6 \cdots \left(\frac{a}{\pi_k}\right)_6.$$

- Moreover $\left(\frac{a^6}{b}\right)_6 = 1$, and

$$\left(\frac{a}{b}\right)_6 = \left(\frac{b}{a}\right)_6 \quad \text{for } a, b \equiv 1 \pmod{12} \quad \longleftarrow \text{ deeply arithmetic}$$

and

$$\left(\frac{ab}{c}\right)_6 = \left(\frac{a}{c}\right)_6 \left(\frac{b}{c}\right)_6.$$

A second attempt at amplifying

So let's make a 2nd attempt at amplifying:

$$H^{1/6}\Big|\sum_{|n|^2\le X}\mu(n)\Big|^2 \le \sum_{|h|^2\le H}\Big|\sum_{|n|^2\le X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2$$

and this comes simply from $\left(\frac{r^6}{n}\right)_6 = 1$, and there being $\approx H^{1/6}$ sixth powers up to H.

The plan

Now the goal is two-fold. Similarly to the twist by $|n|^{2it}$:

1. Transform
$$\sum_{|h|^2 \leq H} \left| \sum_{|n|^2 \leq X} \mu(n) \left(\frac{h}{n}\right)_6 \right|^2$$
as far as harmonic analysis allows: orthogonality in h, then whatever else is available in n.
2. Bound what is left.

First transformation: Poisson in h

Open the square, and apply Poisson in h to each pair n_1, n_2 (in sketch form):

$$\sum_{|h|^2\le H}\Big|\sum_{|n|^2\sim X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2=\sum_{n_1,n_2}\mu(n_1)\mu(n_2)\sum_{|h|^2\le H}\left(\frac{h}{n_1}\right)_6\overline{\left(\frac{h}{n_2}\right)_6}$$

$$\overset{\text{Poisson}}{\approx} HX+\frac{H}{X}\sum_{|k|^2\sim X^2/H}\sum_{n_1,n_2}\mu(n_1)\mu(n_2)\,\overline{\gamma_{-1}(n_1)}\,\gamma_{-1}(n_2)\,\overline{\left(\frac{k}{n_1}\right)_6}\left(\frac{k}{n_2}\right)_6$$

$$=HX+\frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\mu(n)\gamma_{-1}(n)\left(\frac{k}{n}\right)_6\Big|^2$$

where $\gamma_{-1}(n)=\frac{1}{|n|}\sum_{x \bmod n}\overline{\left(\frac{x}{n}\right)_6}\,e\left(\frac{x}{n}\right)$ is a sextic Gauss sum, $e(x)=\exp\left(\frac{4\pi i}{\sqrt{3}}\operatorname{Im} x\right)$.

This transformation does not use anything about μ. If $H \ggg X$ it is beneficial.

So far

Thus we have so far

$$H^{1/6}\Big|\sum_{|n|^2\le X}\mu(n)\Big|^2 \le \sum_{|h|^2\le H}\Big|\sum_{|n|^2\le X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2$$

$$\overset{\text{Poisson (generic)}}{\approx} HX + \frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\mu(n)\gamma_{-1}(n)\left(\frac{k}{n}\right)_6\Big|^2$$

A remarkable identity

Now there is a remarkable identity (for n squarefree, $n \equiv 1 \bmod 12$)

$$\mu(n)\gamma_{-1}(n) = \frac{|n|}{n}\tilde{g}(n)$$

where

$$\tilde{g}(n) = \frac{1}{|n|}\sum_{x \bmod n}\left(\frac{x}{n}\right)_3 e\left(\frac{x}{n}\right)$$

is a cubic Gauss sum.

To verify it one only needs to check it at primes; the rest follows from multiplicativity of μ and the Chinese remainder theorem for the exponential sums.

This has 2 effects:

1. Möbius disappears!
2. This will allow for a 2nd transformation of the sum, using that $\tilde{g}(n)$ is the coefficient of an automorphic form.

So far

Thus we have so far

$$H^{1/6}\Big|\sum_{|n|^2\le X}\mu(n)\Big|^2 \le \sum_{|h|^2\le H}\Big|\sum_{|n|^2\le X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2$$

$$\overset{\text{Poisson (generic)}}{\approx} HX + \frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\mu(n)\gamma_{-1}(n)\left(\frac{k}{n}\right)_6\Big|^2$$

$$\overset{\text{identity for } \gamma_{-1}}{=} HX + \frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\tilde{g}(n)\frac{|n|}{n}\left(\frac{k}{n}\right)_6\Big|^2$$

An automorphic form

Now it turns out that $\tilde{g}(n)$ is the coefficient of an automorphic form (following work of Selberg, Kubota, Patterson),

and this means 2 things:

1. We have a striking formula that shows

   average of twists of Möbius = average of twists of automorphic coefficients

2. We have a 2nd transformation for the sum over n.

Second transformation: Voronoi

Specifically, this follows from the Voronoi formula in my paper with Dunn:

$$\sum_{|n|^2 \sim X} \tilde{g}(n) \frac{|n|}{n} \left(\frac{k}{n}\right)_6 \approx \frac{X}{|k|^2} \sum_{|m|^2 \sim |k|^4/X} b(m) \left(\frac{k}{m}\right)_2$$

for some dual coefficients $b(m)$ with $|b(m)| = 1$ (essentially).

This depends on the construction of a cubic theta function due to Patterson (building on Kubota).

A striking (but known) aspect is that the sextic character becomes a quadratic one. This is the effect of the Kubota multiplier for the cubic theta function.

Combining

Thus combining these identities we arrive at

$$H^{1/6}\Big|\sum_{|n|^2\le X}\mu(n)\Big|^2 \le \sum_{|h|^2\sim H}\Big|\sum_{|n|^2\sim X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2$$

$$\overset{\text{Poisson (generic)}}{\approx} XH+\frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\mu(n)\gamma_{-1}(n)\left(\frac{k}{n}\right)_6\Big|^2$$

$$\overset{\text{identity for } \gamma_{-1}}{=} XH+\frac{H}{X}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|n|^2\sim X}\tilde{g}(n)\frac{|n|}{n}\left(\frac{k}{n}\right)_6\Big|^2$$

$$\overset{\text{automorphy of cubic theta}}{\approx} XH+\frac{H^3}{X^3}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|m|^2\sim X^3/H^2}b(m)\left(\frac{k}{m}\right)_2\Big|^2$$

Note the length dropping (from X to X^3/H^2) if $H \ggg X$, and the sextic character becoming a quadratic character.

The coup de grâce

Finally comes the coup de grâce. For generic coefficients $b(m)$ we have (m, k squarefree)

$$\sum_{|k|^2\sim K}\Big|\sum_{|m|^2\sim M} b(m)\left(\frac{k}{m}\right)_2\Big|^2 \ll_\varepsilon (MK)^\varepsilon(M+K)\sum_{|m|^2\sim M}|b(m)|^2.$$

This (optimal) estimate is due to Heath-Brown (and extended to $\mathbb{Z}[\omega]$ by Goldmakher–Louvel). It is established by iterating TT^*, Poisson and quadratic reciprocity.

Analogues for higher order characters exist, but they are necessarily less efficient (Dunn–R. for cubic, under GRH; de Faveri–Dunn–Hoffstein, cubic and quartic; de Faveri: all orders, conjecturally sharp).

Putting everything together

Thus combining everything we get

$$H^{1/6}\Big|\sum_{|n|^2\le X}\mu(n)\Big|^2 \le \sum_{|h|^2\le H}\Big|\sum_{|n|^2\sim X}\mu(n)\left(\frac{h}{n}\right)_6\Big|^2$$

$$\overset{\text{Poisson}+\gamma_{-1}\text{ identity}\atop{+\text{ automorphy}}}{\approx} XH + \frac{H^3}{X^3}\sum_{|k|^2\sim X^2/H}\Big|\sum_{|m|^2\sim X^3/H^2} b(m)\left(\frac{k}{m}\right)_2\Big|^2$$

$$\overset{\text{H-B large sieve}}{\ll} XH + \frac{H^3}{X^3}\Big(\frac{X^2}{H}+\frac{X^3}{H^2}\Big)\frac{X^3}{H^2}$$

$$= XH + X^2 + \frac{X^3}{H} \ll XH \qquad \text{for } H \gg X.$$

Pick $H = X$ (really $X^{1+\varepsilon}$): then $\left|\sum_{|n|^2\le X}\mu(n)\right|^2 \ll XH^{5/6} = X^{11/6}$, so

$$\sum_{|n|^2\le X}\mu(n) \ll X^{1-\frac{1}{12}} \quad \text{and} \quad \zeta(s)\neq 0 \text{ for } \operatorname{Re} s > \tfrac{11}{12}.$$

A few comments

- Rigorous argument needs to insert cubes in Voronoi (the theta coefficients live on all nb^3, while μ only sees squarefree n). OpenAI handles this with a recursive estimate which takes up a substantial part of their paper; it is possible to get around this using a clever idea of Mayank Pandey (personal communication).
- The drop from $\left(\frac{h}{n}\right)_6$ to $\left(\frac{h}{n}\right)_2$ is a huge efficiency gain for the large sieve.
- The approach parallels some ideas of Iwaniec, roughly spelled out by Conrey in a 2003 Notices article.

The proof in one line

The proof could have been telegraphed. In fact, this is how I remember the argument:

Amplify Möbius in $\mathbb{Z}[\omega]$ using sextic characters.

Apply Poisson, then Voronoi, then the quadratic large sieve.

What is the new idea?

- Looking at Dedekind zeta functions to say something about individual L-functions: known (Deuring–Heilbronn phenomenon, Chebotarev-type results).
- Amplification: standard (Duke–Friedlander–Iwaniec; roots in Weyl/van der Corput).
- The overall strategy of Poisson, Voronoi, ... until the sum is maximally reduced, then the large sieve: very standard.
- Exploiting Gauss sum identities that cancel out Möbius: clever; the identities were known, but few expected them to be helpful.
- The idea that all of this can be combined to obtain quasi-RH: seems a stretch, so nobody realistically thought it could work. The closest precursor is Iwaniec (1998/2002): $\mathbb{Z}[\omega]$, CM curves, Möbius as a sign, but no cubic-theta Voronoi and no quadratic large sieve.